

Gunhee Shin

PhD Student, Urban Robotics Lab @ KAIST

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Daejeon, South Korea

EDUCATION

Ph.D. Student, Electrical Engineering — KAIST (Korea Advanced Institute of Science and Technology) 2025 – Present

- Advisor: Prof. Hyun Myung — Urban Robotics Lab

M.S., Electrical Engineering — KAIST (Korea Advanced Institute of Science and Technology) 2023 – 2025

B.S., Electrical Engineering — Inha University 2017 – 2023

RESEARCH INTERESTS

My research builds robust state estimation, LiDAR-inertial odometry, and SLAM algorithms that let mobile robots and autonomous vehicles operate reliably in real-world environments.

EXPERIENCE

Research Intern — Urban Robotics Lab, KAIST Apr. 2022 – Dec. 2022

- Developed a target-based camera–LiDAR extrinsic calibration pipeline.
- Worked on radar odometry for robust ego-motion estimation.

PUBLICATIONS

JOURNAL ARTICLES

1. **G. Shin**, J. Choi, and K. Kim. "A Fast Correspondence Matching for Iterative Closest Point Algorithm." *The Journal of Korea Robotics Society*, vol. 17, no. 3, pp. 373–380, 2022.

CONFERENCE PAPERS

1. **G. Shin**, S. Lee, J. Kong, Y.-W. Seo, and H. Myung. "SE(3)-LIO: Smooth IMU Propagation With Jointly Distributed Poses on SE(3) Manifold for Accurate and Robust LiDAR-Inertial Odometry." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [arXiv:2603.16118](#) · [project page](#)
2. H. Lim, D. Kim, **G. Shin**, J. Shi, I. Vizzo, H. Myung, J. Park, and L. Carlone. "KISS-Matcher: Fast and Robust Point Cloud Registration Revisited." *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [arXiv:2409.15615](#)
3. **G. Shin**, S. Lee, M. Oh, D. Lee, J. Lee, Y. Seo, and H. Myung. "NM-LIO: Multiple LiDAR-Inertial Odometry Addressing LiDAR Measurement Noise Discrepancy." *International Conference on Robot Intelligence Technology and Applications (RiTA)*, 2024.
4. M. Oh, **G. Shin**, S. Jang, S. Lee, D. Lee, W. Song, B. Yu, H. Lim, J. Lee, and H. Myung. "B-TMS: Bayesian Traversable Terrain Modeling and Segmentation Across 3D LiDAR Scans and Maps for Enhanced Off-Road Navigation." *IEEE Intelligent Vehicles Symposium (IV) Workshop*, 2024. [arXiv:2406.18138](#)
5. H. Lim, K. Han, **G. Shin**, G. Kim, S. Hong, and H. Myung. "ORORA: Outlier-Robust Radar Odometry." *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

PROJECTS

Targetless Multi-Sensor Calibration & SCLAM — Team Lead

2025.06 – 2027.05

- Targetless online LiDAR–camera–radar calibration with calibration-quality monitoring, and SCLAM (Simultaneous Calibration, Localization, and Mapping).

GNSS-Denied Localization & SLAM in Unstructured Environments

2024.08 – 2026.07

- Jamming-robust localization via satellite-image-based global pose estimation, and robust SLAM in unstructured environments.

Robust Perception in Off-Road Terrain

2024.08 – 2025.02

- Environment-change-robust perception and unstructured / off-road terrain information extraction.

Semantic SLAM for Spatial AI

2023.04 – 2023.12

- Benchmarked and evaluated state-of-the-art semantic SLAM (e.g., Hydra) in real-world environments and proposed improvement directions for spatial-AI applications.

SLAM & Perception for Autonomous Legged Robots

2022.08 – 2027.12

- SLAM and object detection in dynamic, unstructured environments for autonomous multi-robot legged locomotion.

Outdoor Multi-Sensor Fusion SLAM & Navigation

2022.03 – 2022.12

- Camera–LiDAR–IMU fusion SLAM with real-time path planning and traversability mapping for outdoor ground-robot navigation (target: under 0.5 m translation error over a 1 km drive).

SKILLS

Languages	C++, Python
Frameworks	ROS, Eigen
Tools	Docker, Git, CMake

HONORS & AWARDS

- 1st Place, Nothing Stands Still (NSS) Challenge, CVPR 2026, Denver, USA — responsible for multi-way registration
- 2nd Place, 3rd Quadruped Robot Challenge (QRC), IEEE ICRA 2025, Atlanta, USA — responsible for pose estimation
- 2nd Place, 2nd Quadruped Robot Challenge (QRC), IEEE ICRA 2024, Yokohama, Japan — responsible for pose estimation
- 1st Place, LiDAR Single-Session — HILTI SLAM Challenge, IEEE ICRA 2023 (1st of 63 teams, with AdaLIO: Adaptive LiDAR-Inertial Odometry)